

Original Manuscript ID: Access-2026-38801

Original Article Title: “The Price of De-correlation: Quantifying the Agreement-Cost Trade-off in Heterogeneous Large Language Model Ensembles”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers’ comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and (c) a clean updated manuscript without highlights (“Main Manuscript”).

The revision adds two substantial pieces of evidence. First, we now measure the correlation between agents’ errors directly rather than inferring it from label agreement, which is what the term “de-correlation” in our title asserts. Second, we ran a pre-registered replication of the entire protocol on a substantially different domain, 537 expert clinical-decision items, and report it whatever the outcome. Every number below is regenerated by the released code from the frozen data.

Best regards,

Samir Chincholikar
Robin Chawla

Reviewer #1, Concern #1

Reviewer comment:

Reduced agreement is not shown to improve accuracy or any decision-relevant outcome. Link the agreement-cost frontier to calibration, abstention, robustness, or tail-error reduction.

Author response:

- We agree, and we have not claimed otherwise. The manuscript reports that heterogeneity produces no reliable gain in average directional accuracy; the reviewer is right that this leaves the frontier without a demonstrated downstream benefit.
- We now test the most directly relevant of the four suggested outcomes, abstention, using a selective-prediction analysis, and we report the result even though it is null.
- Ensemble decisions are ranked by mean agent conviction, coverage is swept from the most confident decision to the full set, and selective risk is the error rate among retained decisions. The area under the risk-coverage curve is 0.553 for HOM, 0.568 for HET-LITE and 0.554 for HET, against full-coverage risks of 0.562, 0.537 and 0.512. Restricting to the most confident half of decisions does not materially reduce risk in any configuration.
- Conviction as elicited here does not rank errors usefully, so we claim no calibration or abstention benefit.
- We do report one decision-relevant quantity that does move: the rate at which an ensemble is unanimous and wrong. See our response to Concern #4.

Author action:

- Added “Selective prediction” to the results, reporting the risk-coverage analysis and AURC for all three configurations.
- The frontier is now stated explicitly as a description of what diversity costs, not as evidence that it improves decision quality.

Reviewer #1, Concern #2

Reviewer comment:

The tier-matched control covers only two providers and no xAI equivalent. A fully tier-matched multi-provider design or model-level random effects is needed.

Author response:

- The observation is correct: the capability-matched control in the original submission used OpenAI and Google only.
- The reviewer offers two remedies. The cross-domain replication described under Concern #3 supplies the first in a form we can report with confidence, because on that domain two of the three providers have near-identical accuracy.
- Within the fully heterogeneous configuration we compare agent pairs whose two members have near-identical accuracy but differ in whether they share a provider. The same-provider pairing and the cross-provider pairing answer those items at 0.951 and 0.955 mean member accuracy respectively, a gap of 0.005. The same-provider Gemini pair has error correlation 0.954; the cross-provider Gemini-Grok pairing has 0.607, a difference of 0.347 (95% CI [0.207, 0.507]).

- The comparison holds the item set, the configuration and the ensemble composition fixed, and the two pairings have near-identical observed member accuracy (0.951 and 0.955). We describe this as approximate capability matching, which is the wording the manuscript uses; it is not matching by construction.
- Every same-provider pairing shows higher error correlation than every cross-provider pairing, at both accuracy levels present in the ensemble.

Author action:

- Added the capability-matched provider contrast to the cross-domain section.
- Stated in that section that the provider-associated ordering persists after approximate capability matching, while provider remains confounded with other model-specific characteristics and is not isolated causally.

Reviewer #1, Concern #3

Reviewer comment:

Findings come from one financial task, one prompt, three labels, and fixed five-agent ensembles. Test other domains, ensemble sizes, aggregation rules, and label structures.

Author response:

- We have run the replication the reviewer asks for, on a domain chosen to be as unlike the original as possible: expert clinical decision-making, using the MMLU professional-medicine and clinical-knowledge subsets, 537 four-option items under the MIT licence with no personal data.
- The protocol is otherwise unchanged: the same five agents, the same three configurations, the same per-agent independent-draw seeding, the same temperature, the same number of runs and the same estimators. The domain is what the replication varies; the response structure also changes, from three ordered labels to four options with one correct answer, which is inherent in moving to a knowledge task and which we state in the manuscript. No other element of the protocol was redesigned.
- The design, the predicted direction and the decision rule were registered and hashed before collection began.
- We predicted in advance that the effect would replicate in direction but be smaller than in finance, because agents answer these items far more accurately and agents that are more often correct necessarily agree more. Both parts hold. Agreement falls by 0.072 (95% CI [0.040, 0.104]), about a fifth of the financial estimate of 0.336, while error correlation falls by 0.393 (95% CI [0.325, 0.459]), close to the financial estimate of 0.454.
- This also changes the label structure the reviewer asks about: four categories rather than three.
- We have not varied ensemble size or aggregation rule. Those are single-parameter sweeps on the frozen data rather than new evidence about generality, and we preferred to spend the revision on the domain question, which is the one that bears on whether the finding is a property of the task. We state that omission rather than leaving it unexplained.

Author action:

- Added “Cross-domain replication: clinical decisions” to the results.
- Added the pre-registration and freeze receipt for that arm to the released artifact.

Reviewer #1, Concern #4

Reviewer comment:

The 1.84x unanimous-incorrect finding and HET-LITE recommendation are post hoc, lack CIs/p-values, and ignore multiplicity. Treat as hypothesis-generating only.

Author response:

- We accept the framing and have adopted the remedy the reviewer specifies.
- The interval is now reported. The ratio is 1.84 with a 95% equity-clustered bootstrap confidence interval of [1.42, 2.52], which excludes unity. The underlying rates are 0.0583 for HOM and 0.0317 for HET over 3600 scoreable cells each.
- This interval has been corrected since the previous revision, which previously reported the superseded [1.51, 2.35]. That bootstrap drew a SET of equities, so an equity drawn twice counted once and each draw used only the distinct equities of a 100-draw multiset. That is subsampling rather than a cluster bootstrap and it understated the variance. It is the same defect the reviewer identified in the clustering table, found by auditing for it elsewhere once it had been corrected there.
- Because this statement is inferential rather than descriptive, it also carries the two-way sensitivity: admitting date dependence widens the interval to [1.13, 3.06], which still excludes unity.
- We state plainly that the configuration compared and the statistic reported were both chosen after inspecting the data, that several such comparisons are available within that section, and that no multiplicity adjustment is applied.
- We have withdrawn HET-LITE as a design recommendation. It is now recorded as a hypothesis for a future confirmatory study, not as guidance for deployment.

Author action:

- Added the confidence interval and an explicit multiplicity caveat to the exploratory section.
- Removed the recommendation language.

Reviewer #1, Concern #5

Reviewer comment:

Seed support is best-effort, and the diagnostic is strongest only for HOM. Heterogeneous cross-run checks are confounded by repeated same-model draws. Add verification or sensitivity analysis.

Author response:

- Both observations are correct, and we have added the sensitivity analysis rather than claiming a verification the provider APIs do not support.
- Decomposing pairwise agreement by whether the two agents share a model separates the two effects the reviewer identifies. Same-model pairs agree at 0.800 in HOM, 0.794 in HET-LITE and 0.862 in HET. Different-model pairs agree at 0.455 and 0.490 respectively.
- Same-model agreement is stable across configurations. The fall in kappa is therefore driven by the introduction of different-model pairs, not by any change in how same-model pairs behave, which is the confound the reviewer describes.
- We also now state what unique seeds do and do not establish. A distinct seed removes one specific

artifact, identical sampling trajectories from a shared seed. It does not establish statistical independence: commercial inference stacks do not guarantee bitwise-reproducible outputs for a fixed seed, because reduction order varies with kernel selection and batch composition.

Author action:

- Added “Sensitivity: agreement by pair type” and “What unique seeds do and do not establish”.

Reviewer #1, Concern #6

Reviewer comment:

Table 4 and the text disagree on costs. Clarify per-decision cost, token accounting, and pricing assumptions.

Author response:

- We have checked this carefully and do not find a disagreement. Table 4 gives 6.48×10^{-4} , 1.52×10^{-3} and 2.84×10^{-3} per ensemble decision; the text gives the same three quantities rounded to two significant figures, 6.5×10^{-4} , 1.5×10^{-3} and 2.8×10^{-3} ; and the ratio $2.835/0.648 = 4.375$ is the 4.4x quoted. The released analysis regenerates all three values to six decimal places from the metered token counts.
- We think the substantive point behind the observation is that the manuscript never stated how per-decision cost is computed, so the figures could not be checked. That was an omission and we have corrected it.

Author action:

- Added the cost basis to the frontier section: metered prompt and completion tokens priced at the per-million rates in the model registry, summed over the five agents in a cell and averaged over cells; prices fixed across configurations; no discount, batching or caching assumed; failed calls excluded because they return no billable completion.

Reviewer #2, Concern #1

Reviewer comment:

Simplify index terms to 3 - 5 words only.

Author response:

- Done. The list has been reduced from eight terms to five.

Author action:

- Index terms are now: large language models, multi-agent systems, model heterogeneity, inter-rater agreement, algorithmic monoculture.

Reviewer #2, Concern #2

Reviewer comment:

No clear explanation about fair comparison with current state-of-the-art approach.

Author response:

- The manuscript did not explain why it contains no head-to-head benchmark, and that was a gap.
- This article does not propose an ensemble method. Methods such as mixture-of-agents, multi-agent debate and diversity-maximising ensemble selection are evaluated on task performance, and the comparison that matters for them is accuracy against a baseline. The object of study here is different: how much behavioral de-correlation a given ensemble composition buys and what it costs, holding task, prompt and information set fixed.
- The fair comparison is therefore one of measurement. Prior work reporting ensemble diversity typically uses a single agreement index on pooled outputs, without separating agreement from error correlation, without an independent-draw protocol, and without interval estimates. The manuscript now shows that the first distinction changes the interpretation, the second changes the estimate roughly fourfold in our pilot, and the third determines whether a reported difference is resolvable at all.

Author action:

- Added “Relation to state-of-the-art ensemble methods”.

Reviewer #3, Concern #1

Reviewer comment:

The present experiment confounds provider, model identity, capability, training data, architecture, and alignment. Add same-provider/different-model controls, more than one model per provider, balanced provider mixtures, and preferably a factorial analysis; otherwise rewrite all causal/provider-specific claims as configuration-specific observational findings.

Author response:

- The confound is real and we have taken the second of the two courses the reviewer offers, while also supplying the capability-matched evidence described under Reviewer #1, Concern #2.
- Provider is not experimentally separable from model identity, parameter count, training corpus, architecture or alignment procedure in this design, because each provider contributes one model. We now say so directly, and state that “provider” should be read throughout as shorthand for that bundle rather than as an isolated causal factor.
- The cross-domain arm adds a capability-matched version of the comparison in which the two members of each pairing have near-identical accuracy, and provider continues to order the result there. That narrows the confound without claiming to eliminate it.
- We state what design would be required to decompose the bundle further, and that it is not attempted here.

Author action:

- Rewrote the provider-pair discussion as a configuration-specific observational finding.
- Added the capability-matched contrast in the cross-domain section.

Reviewer #3, Concern #2

Reviewer comment:

Recompute uncertainty while accounting for both repeated observations within equities and shared analysis dates/common market conditions. Justify the bootstrap unit formally, evaluate two-way equity/date clustering

or an appropriate multiway resampling strategy, and regenerate all main and exploratory confidence intervals accordingly.

Author response:

- We have done this, and the primary result survives it.
- The equity is the pre-registered unit because it is the level at which repeated observations are nested: each security is evaluated on twelve dates and shares company-specific context. We now state that justification rather than assuming it.
- Dates are a second, crossed source of dependence. We report the primary contrast under three schemes: equity clustering (100 clusters), date clustering (12 clusters), and a two-way scheme in which equities and dates are resampled independently.
- The point estimate is 0.3363 in all three. The intervals are [0.3035, 0.3689] for equities, [0.2989, 0.3656] for dates and [0.2786, 0.3877] two-way. The two-way interval is the widest, as expected when two crossed dependence structures are admitted at once, and all three exclude zero.
- The reviewer is right that the earlier equity interval, [0.3131, 0.3628], contradicted the primary [0.3035, 0.3689] reported in the abstract, Table 2, the Results and the Conclusion. They were presented as the same estimand and should have agreed exactly. They did not because the two were computed by different procedures: the primary bootstrap resamples equities with replacement and counts an equity drawn twice as two clusters, while the robustness table treated each draw as a set, so a repeated draw counted once and every draw used only the roughly 63 distinct equities of a 100-draw multiset. That is subsampling, not a cluster bootstrap, and it understated the variance. The robustness table now honours multiplicity and uses the pre-registered equity ordering, so its equity row reproduces the primary interval exactly; the code asserts that equality on every run and fails if it is ever broken. The date and two-way intervals widen for the same reason. No point estimate changed and every interval still excludes zero.
- We note that the date margin has only twelve clusters, a regime in which cluster-robust inference is known to under-cover. The equity-clustered interval therefore remains the pre-registered primary and the other two are reported as sensitivity analyses.
- On scope, the reviewer asked that all main and exploratory intervals account for equity and date dependence. The two-way scheme is applied to the primary endpoint and, in this revision, to the two exploratory contrasts that carry an inferential claim: the pile-on contrast widens from [0.160, 0.216] to [0.140, 0.232] and still excludes zero, and the AURC contrast widens from [-0.059, 0.060] to [-0.113, 0.115] and still contains zero. Neither conclusion changes. The remaining exploratory quantities are reported as descriptive statistics without an inferential claim attached, and retain the pre-registered paired equity-clustered bootstrap; they are interpreted as exploratory throughout and we do not draw conclusions from them that a wider interval could overturn.

Author action:

- Added the bootstrap-unit justification to the uncertainty section.
- Added a table reporting the primary contrast under all three resampling schemes.

Reviewer #3, Concern #3

Reviewer comment:

Fleiss' kappa measures agreement, not correlation. Either remove “de-correlation”/“less-correlated judgments” throughout the title, abstract, figures, discussion, and conclusion, or add actual dependence/error-correlation

analyses that justify those terms. Explicitly distinguish output agreement, error correlation, sampling independence, model diversity, and provider diversity.

Author response:

- The reviewer is right that kappa measures agreement irrespective of correctness, and the original manuscript acknowledged the distinction without measuring the second quantity. We have taken the second option and added the analysis.
- For every pair of agents within a cell we compute the Pearson correlation between their binary error indicators against the forward-return sign. Error correlation falls monotonically: 0.984 for HOM, 0.742 for HET-LITE and 0.530 for HET, giving 0.454 (95% CI [0.376, 0.534]).
- Because abstentions are excluded, the scored subset differs across configurations. We repeat the calculation counting HOLD as an error, which fixes an identical subset for every configuration: the contrast is then 0.309 (95% CI [0.262, 0.354]). Both treatments give a monotone decline whose interval excludes zero, so the estimate is bracketed rather than dependent on how abstentions are handled.
- Mean per-agent error rates are close across configurations, so the decline is a change in dependence and not a consequence of differing accuracy.
- We also state that de-correlation is partial rather than complete: heterogeneous agents still err together more often than independence would imply.
- We have added the five-way distinction the reviewer asks for, and we now define the term explicitly as sense (2), error correlation, with the phi contrast rather than the kappa contrast as its evidence.

Author action:

- Added “Agreement, error correlation, and decorrelation”, containing the five-way distinction and the phi analysis under both label treatments. That subsection defines decorrelation as a reduction in error correlation measured by phi, and the sections that report kappa have been retitled accordingly: the primary results section is now “Heterogeneity reduces ensemble agreement”.
- The abstract now reports the phi contrast alongside the kappa contrast.

Reviewer #3, Concern #4

Reviewer comment:

HOLD-induced abstention creates different evaluation subsets across configurations, so the reported hit rates are not directly comparable. Perform a proper selective-prediction/coverage analysis, use paired and clustered inference, provide uncertainty for unanimous-error and pile-on statistics, and stop using overlapping marginal Wilson intervals as the main inferential evidence.

Author response:

- We agree on all four points. Three were acted on in the previous revision; the fourth, uncertainty for the pile-on statistic, was claimed but not delivered, and the reviewer is correct to press it. It is now provided.
- The selective-prediction and coverage analysis is described under Reviewer #1, Concern #1.
- Uncertainty for the unanimous-error statistic is described under Reviewer #1, Concern #4; the interval is equity-clustered rather than marginal.
- Pile-on now carries uncertainty. The rate at which the remaining agents endorse an incorrect majority is 0.824 for HOM and 0.636 for HET; a paired equity-clustered bootstrap over the same 100 equities, 2,000

draws, using the resampling scheme of the primary endpoint, gives a HOM-HET difference of 0.189 (95% CI [0.160, 0.216]). The draw is shared between the two arms and the difference taken within it, because the two configurations are measured on the same equities and resampling them independently would add back the between-arm variance the design removes.

- Paired clustered uncertainty for AURC is also feasible and is now reported: the HOM-HET difference is -0.001 (95% CI [-0.059, 0.060]). The interval contains zero, so these data do not establish a difference in selective-prediction quality between the two configurations, and the manuscript now says exactly that rather than comparing the three AURC values informally.
- The error-correlation analysis under Reviewer #3, Concern #3 is reported under two label treatments precisely because of the subset problem the reviewer identifies: the second treatment fixes an identical subset across configurations and removes the selection.
- Wilson intervals are no longer the main inferential evidence anywhere. They appear once, in the secondary accuracy table, and the inference that section relies on is now the clustered contrast.

Author action:

- Added the selective-prediction analysis and the clustered interval on the unanimous-error ratio.
- Added paired equity-clustered bootstrap confidence intervals for the pile-on contrast and for the AURC contrast, and reported both in the Results.
- Reported the error-correlation contrast under both label treatments.

Reviewer #3, Concern #5

Reviewer comment:

Clarify that unique seeds do not prove statistical independence; provide complete API parameters and versions, exact prompts/system messages, seed-support evidence, retry records, total attempted versus successful API calls, exact preregistration timestamp evidence, and archived immutable code/data. Correct Figure 3 so that the shared-seed pilot, independent-seed pilot, and full-study quantities compare the same estimand.

Author response:

- The seed clarification is described under Reviewer #1, Concern #5.
- On provenance: the confirmatory grid comprises 54,000 planned calls, all of which returned a valid response. Reaching them required 94,566 API requests. Of the 40,566 that did not return a usable completion, 40,544 were provider rate-limit or quota responses and 22 were transport or schema failures. A rate-limited request is rejected before the model is invoked and returns no content, so retrying it cannot select on the response; the completed grid is the full pre-registered design, not a subset surviving a content filter.
- Request parameters, prompt text, per-row prompt hashes, model snapshot identifiers, prices and cutoffs are all released, and every attempt including failures is in the log with its error string.
- On Figure 3 the reviewer is correct and we had conflated two quantities. Two of the bars were the pilot's HOM-HET contrast under shared and independent seeding on the same grid and model set. The third was the confirmatory study's within-minus-cross-run difference for HOM, which is a residual-dependence diagnostic and not a HOM-HET contrast at all.
- We have done exactly what the reviewer asked. Figure 3 now contains only the two like-for-like pilot quantities, so every bar in it measures the same estimand: Delta-kappa(HOM-HET) on the same 8-equity

pilot grid and the same model set, differing only in whether the five agents in a cell share one seed (0.485) or draw independent seeds (0.113). The full-study within-minus-cross-run diagnostic has been removed from the figure and is reported separately in Table 6, where it is identified as a diagnostic rather than a contrast.

- An earlier revision kept all three bars and relabelled them instead. That was not what the reviewer requested and we have not retained it.

Author action:

- Added “Collection provenance and retry accounting”, reporting attempted versus successful calls, the failure breakdown and the fixed request parameters.
- Reduced Figure 3 to the two like-for-like pilot bars and moved the full-study within-minus-cross-run diagnostic to Table 6; rewrote the caption accordingly.

Reviewer #3, Concern #6

Reviewer comment:

Replicate the main experiment on at least one substantially different domain/task and preferably multiple prompt formats, ensemble sizes, and model mixtures. The present single BUY/HOLD/SELL financial task-whose accuracy is itself near chance and whose label marginals vary strongly by configuration-is insufficient to establish a general principle about heterogeneous LLM ensembles.

Author response:

- We have run the replication, and we chose the domain to answer the two specific criticisms in the reviewer’s sentence as well as the general one.
- On near-chance accuracy: chance on the new task is 25% and agents answer well above it.
- On label marginals: the four options are balanced by construction. The published answer key is skewed toward one position, so choice order is shuffled deterministically per item, which we document and release.
- The result is reported under Reviewer #1, Concern #3. Both pre-registered predictions hold.
- The comparison across the two domains is itself informative. Agreement compresses by roughly a factor of five between domains while error correlation barely attenuates (0.393 against 0.454). Error correlation attenuates substantially less than agreement across the two tested domains, which is why the de-correlation claim is now stated in terms of ϕ . Two domains do not establish portability in general, and we do not claim it.
- We have not varied prompt format or ensemble size, and we say so. Those are the “preferably” items in the reviewer’s request; we judged the domain replication to be the part that bears on generality and preferred to do one thing completely.

Author action:

- Added the cross-domain replication described above, with its pre-registration and freeze receipt in the released artifact.